

School of Science, Computing and Emerging
Technologies



Unit Outline

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COS40007

Artificial Intelligence Engineering

Semester September 2026

Please read this Unit Outline carefully. It includes:

- PART A** Unit summary
- PART B** Your Unit in more detail
- PART C** Further information



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PART A: Unit Summary

Unit Code(s)		COS40007
Unit Title		Artificial Intelligence Engineering
Duration		One Semester
Total Contact Hours		150 Hours
Requisites:		
	Pre-requisites	100 cp in BEng or BCompSc or related double degrees AND COS10009 Introduction to Programming
	Co-requisites	NIL
	Concurrent prerequisites	NIL
	Anti-requisites	NIL
	Assumed knowledge	NIL
Credit Points		12.5
Campus/Location		Swinburne Vietnam
Mode of Delivery		Online seminar and on-campus workshop
Assessment Summary		Applied Project (Group): 52% Portfolio (Individual): 48% Important: Check Hurdle Requirements under Assessment Requirements

Aims

This unit of study aims to equip engineers with both theoretical knowledge and practical skills to design, implement, and evaluate Artificial Intelligence (AI) and Machine Learning (ML) solutions for complex engineering challenges. It emphasises developing proficiency in selecting suitable algorithms, optimising system performance, and considering ethical aspects, while bridging the gap between Artificial Intelligence (AI) theory and engineering applications. The course prepares professionals to deploy intelligent systems efficiently and communicate technical solutions across multidisciplinary teams.

Unit Learning Outcomes

Students who successfully complete this unit can:

1. Design, develop, deploy, and maintain Machine Learning (ML)/Artificial Intelligence (AI) pipelines by selecting and training datasets to address complex, multidisciplinary engineering problems.
2. Explain and compare a range of Artificial Intelligence (AI), Machine Learning (ML), & Deep Learning (DL) algorithms in terms of their structure, function, and real-world applicability.
3. Evaluate and justify the selection of AI techniques based on computational efficiency, ethical implications, and problem constraints for solving diverse computational engineering challenges.
4. Communicate clearly and effectively technical concepts, project outcomes, and design decisions through written reports and oral presentations.

Graduate Attributes

The Swinburne Graduate Attributes describe the capability of our graduates to use knowledge, skills and behaviours to contribute to society meaningfully and positively. They include professional, self-directed learning and future-ready skills.

This unit contributes to the development of the following Swinburne Graduate Attributes:

- GA1 Communication - Verbal communication
- GA2 Communication - Communicating using different media
- GA3 Teamwork - Collaboration and negotiation
- GA4 Teamwork – Teamwork roles and processes
- GA5 Digital literacies– Information literacy
- GA6 Digital Literacies– Technical literacy

Other graduate attributes may be practised in the unit but are not formally taught as part of the unit content, nor incorporated within formal assessment.

Content

- Utilising Machine Learning (ML) paradigms such as Supervised, Unsupervised, and Reinforcement Learning.
- Understanding foundation models vs traditional machine learning trade-offs.
- Data-Centric Artificial Intelligence (AI) Development.
- Automated Data Pipelines and Machine Learning Operations (MLOps) for reproducibility.
- Algorithm Selection Frameworks for Production-Grade Machine Learning (ML) Specifications.
- Artificial Intelligence (AI) for Emerging Engineering Applications & Technologies.

PART B: Your Unit in more detail

Unit Improvements

Feedback provided by previous students through the Student Survey has resulted in improvements that have been made to this unit. Recent improvements include:

- The Unit Content and name change has been updated to reflect the unit's focus on the design, development, and deployment of AI systems within an engineering context. It also aligns with current industry terminology and emerging job roles (e.g., AI Engineer), and more accurately communicates the hands-on, system-oriented nature of the unit content.

Unit Teaching Staff

Name	Role	Location	Email / Teams	Consultation Times
Dr. Mai Duc Tho	Unit Coordinator	Hanoi	tmai@swin.edu.au	By appointment
Mr. Minh Anh Hoang	Lead Coordinator	HCMC	manh@swin.edu.au	By appointment

Learning and Teaching Structure

Category	Activity	Total Hours	Hours per Week	Teaching Period Weeks
Live Online	Lecture	12 hours	1 hours	Weeks 1 to 12
Online	Asynchronous Lecture	12 Hours	1 Hours	Weeks 1 to 12
In-person/On-Campus	Studio	24 hours	2 hours	Weeks 1 to 12
Online	Self-paced online learning activities	102 hours	8.5 hours	Weeks 1 to 12

Week by Week Schedule

Week	Week Beginning	Teaching and Learning Activity	Student Task or Assessment
1.	14 Sep	Seminar: Introduction to the AI Engineering Design Process and AI Model Life Cycle.	
2.	21 Sep	Seminar: Data Engineering	
3.	28 Sep	Seminar: Software Engineering and DevOps for AI	Submission of Portfolio Task 1
4.	5 Oct	Seminar: Building AI Apps: Traditional ML & Feature Engineering	
5.	12 Oct	Seminar: Building AI Apps: Deep Learning & Foundation Models	Submission of Portfolio Task 2
6.	19 Oct	Seminar: Responsible AI: Ethics, Fairness, Bias Mitigation	Submission of Applied Project Task 1
7.	26 Oct	Seminar: System Life Cycle in AI & Project Management	Submission of Portfolio Task 3
8.	2 Nov	Seminar: Deployment and MLOps Practices	Submission of Applied Project Task 2
9.	9 Nov	Seminar: Scaling and Monitoring AI Systems	
10.	16 Nov	Seminar: AI Security & Privacy Preserving Techniques	

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11.	23 Nov	Seminar: AI Trustworthiness with Case Studies on AI for the Engineering Domain.	Submission of Applied Project Task 3
12.	30 Nov	Seminar: Future of AI Engineering (+ Optional Industry Project Talk)	Online Demo/On-Campus Applied Project Presentation
13.	7 Dec		Submission of Final Applied Project Report + Peer Review

Assessment

a) Assessment overview

Tasks and Details	Individual or Group	Weighting	Mapped Unit Learning Outcomes	Mapped Graduate Attributes	Assessment Due Date
1. Portfolio Task	Individual	48%	1,2,3,4	GA1, GA2, GA5, GA6	During the middle of Weeks 3, 5, 7: Individual Portfolio Submissions
2. Applied Project Task	Group	25%	1,2,3,4	GA1, GA2, GA3, GA4, GA5, GA6	During the middle of Weeks 6, 8, 11: Group Project Submissions
3. Video Presentation	Group	5%	All	GA1, GA2	During Week 12: Online /On-Campus
4. Final Applied Project Report	Group	22%	1,2,3,4	GA1, GA2, GA3, GA4, GA5, GA6	End of Week 13
5. Peer Review	Individual	0%	All		End of week 13

*Students can be asked to demo during On-Campus Studios with the tutor's discretion.

* To be determined later.

Assessment Requirements	Details
b) Use of generative AI (genAI) in this unit	Permitted use of genAI varies by assessment task in this unit. Refer to each assessment description below for specific guidance. No AI for Individual Portfolio & Group Assessments: The Portfolio and Group Tasks in this unit should be completed entirely without genAI assistance. Report Writing for Assessments: genAI may be used for brainstorming, structuring, and generating ideas to improve work. Any use of genAI must be acknowledged, with prompts and outputs included in an appendix.

c) Hurdle requirements	To pass this unit, you must: (i) Achieve an overall mark for the unit of 50% or more, and (ii) Complete the project to an acceptable standard. A rubric will be used to determine if students have met the acceptable standard. The rubric is available on Canvas, and (iii) Achieve a minimum of 50% or more on the Portfolio (must pass at least 50% of the portfolio assessment activities). Students who do not successfully achieve hurdle requirements (ii) and (iii) in full will receive a maximum of 45% as the total mark for the unit.
d) Final assessment period	There is no final examination for this unit. However, some submissions are scheduled during the examination period.
e) Submission requirements	Assignments and other assessments are generally submitted online through the Canvas assessment submission system which integrates with the Turnitin. The Assessment Cover Sheet should be submitted with the assignment. The GitHub Repository Link should be submitted with each submission.
	Please ensure you keep a copy of all assessments that are submitted. In cases where a hard copy submission is required an Assessment Cover Sheet must be submitted with your assignment. The standard Assessment Cover Sheet is available from the Submitting work webpage or www.swinburne.edu.au/studentforms/

f) Extensions, late penalties and special consideration	<u>Extensions</u> For extensions (other than those granted through Special Consideration), please contact your Unit Convenor who will assess on a case-by-case basis. <u>Late Penalties</u> Unless an extension has been approved, late submissions will result in a mark penalty. You will be penalised 5% of the available mark for each calendar day the task is late, up to a maximum of seven calendar days. After seven calendar days, a zero mark will be recorded. For example: - if you achieve 90/100 on an assessment task but the task was submitted two days late, a late penalty of 10 is applied, and your mark is recorded as 80/100. - if you achieve 29/35 on an assessment but the task was submitted five days late, a late penalty of 8.75 (25% of the 35 available marks) is applied, and your mark is recorded as 20.25/35. <u>Special Consideration</u> If your studies have been adversely affected by serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you can apply for special consideration. Applications for Special Consideration are made via an online form <u>no later than 5.00pm</u> of the third working day after the assessment due date. For further information about eligibility and how to apply, see https://www.swinburne.edu.au/life-atswinburne/student-support-services/specialconsideration-assistance/
g) Referencing	To avoid breaching academic integrity, you are required to provide references whenever you include information from other sources in your work and acknowledge when you have used Artificial Intelligence (AI) tools (such as Microsoft Copilot). Further details regarding academic integrity are available in Section C of this document. Referencing conventions required for this unit are:
	IEEE citation style (https://pitt.libguides.com/citationhelp/ieee) Helpful information on referencing can be found at http://www.swinburne.edu.au/library/referencing/

h) Groupwork guidelines	A group assignment is the collective responsibility of the entire group, and if one member is temporarily unable to contribute, the group should be able to reallocate responsibilities to keep to schedule. In the event of longer-term illness or other serious problems involving a member of group, it is the responsibility of the other members to notify immediately the Unit Convenor or relevant Tutor. Group submissions must be submitted with an Assignment Cover Sheet, signed by all members of the group. All group members must be satisfied that the work has been correctly submitted. Any penalties for late submission will generally apply to all group members, not just the person who submitted.
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b) Final unit results

a) Supplementary assessment	Supplementary assessment is not available in this unit due to learning verification requirements.
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Required Textbook(s)

The required textbook(s) can be purchased from bookshops and may be available through the Swinburne Library.

1. Huyen, C. (2024). *AI engineering: Building applications with foundation models*. " O'Reilly Media, Inc."
2. Goodfellow, I., Bengio, Y., Courville, A., & Bengio, Y. (2016). *Deep learning* (Vol. 1, No. 2, pp. 1-800). Cambridge: MIT press.
3. Bass, L., Lu, Q., Weber, I., & Zhu, L. (2025). *Engineering AI systems: architecture and DevOps essentials*. Addison-Wesley Professional.
4. Raschka, S., Liu, Y. H., & Mirjalili, V. (2022). *Machine Learning with PyTorch and Scikit-Learn: Develop*
5. *machine learning and deep learning models with Python*. Packt Publishing Ltd.
6. Kapoor, A., Gulli, A., Pal, S., & Chollet, F. (2022). *Deep Learning with TensorFlow and Keras: Build and deploy supervised, unsupervised, deep, and reinforcement learning models*. Packt Publishing Ltd.

Recommended Reading Materials

Swinburne Library has a large collection of resources. Listed below are some references that will provide valuable supplementary information to this unit. It is also recommended that you explore other sources to broaden your understanding.

- https://librarysearch.swinburne.edu.au/permalink/61SUT_INST/7ojd5o/alma99386298_46201361
- https://librarysearch.swinburne.edu.au/permalink/61SUT_INST/7ojd5o/alma9938703980_401361

Readings TAB in Canvas:

https://swinburne.instructure.com/courses/73590/external_tools/6356

PART C: FURTHER INFORMATION

For further information on any of these topics, refer to Swinburne
Student webpage <http://www.swinburne.edu.au/st/udent>

Student behaviour and wellbeing

All students are expected to: act with integrity, honesty and fairness; be inclusive, ethical and respectful of others; and appropriately use University resources, information, equipment and facilities. All students are expected to contribute to creating a work and study environment that is safe and free from bullying, violence, discrimination, sexual harassment, vilification and other forms of unacceptable behaviour.

The [Student Charter](#) describes what students can reasonably expect from Swinburne in order to enjoy a quality learning experience. The Charter also sets out what is expected of students with regards to your studies and the way you conduct yourself towards other people and property.

You are expected to familiarise yourself with University regulations and policies and are obliged to abide by these, including the [Student Academic Misconduct Regulations](#), [Student General Misconduct Regulations](#) and the [People, Culture and Integrity Policy](#). Any student found to be in breach of these may be subject to disciplinary processes.

Examples of expected behaviours are:

- conducting yourself in teaching areas in a manner that is professional and not disruptive to others
- following specific safety procedures in Swinburne laboratories, such as wearing appropriate footwear and safety equipment, not acting in a manner which is dangerous or disruptive (e.g. playing computer games), and not bringing in food or drink
- following emergency and evacuation procedures and following instructions given by staff/wardens in an emergency response.

Canvas

You should regularly log on to the Swinburne learning management system, Canvas. You can access Canvas via the [Student login](#) webpage or <https://swinburne.instructure.com/> Canvas is updated regularly with important unit information and communications.

Communication

All communication will be via your Swinburne email address. If you access your email through a provider other than Swinburne, then it is your responsibility to ensure that your Swinburne email is redirected to your private email address.

Academic Integrity

Academic integrity is about taking responsibility for your learning and submitting work that is honestly your own. It means acknowledging the ideas, contributions and work of others; referencing your sources and acknowledging the use of generative artificial intelligence; contributing fairly to group work; and completing tasks, tests and exams without cheating. Artificial intelligence tools should only be used where approved by the Unit Convenor.

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Swinburne University uses the Turnitin system, which helps to identify inadequate citations, poor paraphrasing and unoriginal work in assignments that are submitted via Canvas. Your Unit Convenor will provide further details.

Plagiarism, collusion, contract cheating, unauthorised file sharing, falsification, fabrication, manipulation or misrepresentation of information, reuse of previous work and noncompliance with instructions in an invigilated or non-invigilated assessment item are all breaches of academic integrity and treated as academic misconduct. Examples of breaches of academic integrity include, but are not limited to:

- submitting work as your own for assessment that has been fully or partially completed by a third party, either paid or unpaid
- using output from artificial intelligence tools (e.g. ChatGPT) in whole or part without acknowledgement and/or without the approval of the Unit Convenor
- using another person's work or ideas as though it is your own work, without appropriate attribution
- working closely with another student or group of students (either past or current), to submit for assessment, some or all of the other student or students' work as your own work
- sharing without permission of the Unit Convenor, Swinburne resources or other material related to assessment to an entity or document repository site
- creating, intentionally modifying or inventing information that is intended to be submitted as part of an assessment item
- using the whole or part of a computer program written by another person as your own without appropriate acknowledgement
- poorly paraphrasing somebody else's work
- using a musical composition or audio, visual, graphic and photographic work created by another person without acknowledgement
- enabling others to cheat, including letting another student copy your work or by giving access to a draft or completed assignment
- letting someone or something else impersonate you, or you impersonate someone else in an invigilated or non-invigilated assessment item
- accessing, obtaining and/or providing to others unauthorised materials relating to an invigilated or non-invigilated assessment item.

The penalties for academic misconduct can be severe, ranging from a zero grade for an assessment task through to exclusion from Swinburne. For further details, see <https://www.swinburne.edu.au/student-login/academic-integrity/>

Student support

Swinburne offers a range of services and resources to help you complete your studies successfully, such as study and learning support and health and wellbeing services. Your Unit Convenor or studentHQ can provide information about the study support and other services available for Swinburne students. For further information, see the [Student support services](#) web page.

Special consideration

If your studies have been adversely affected due to serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you may be able to apply for special consideration (SPC).

Applications for Special Consideration are submitted via the SPC online tool normally no later than 5.00pm on the third working day after the submission/sitting date for the relevant assessment component. See

<https://www.swinburne.edu.au/life-at-swinburne/studentsupport-services/special-consideration-assistance/>

Note: Submitting fraudulent (fake or altered) medical certificates is considered misconduct and can lead to serious penalties from Swinburne. In addition, your doctor may report fraudulent medical certificates as a prosecutable offence under the Victorian Crimes Act.

AccessAbility Services

If you are a student with a disability, medical or mental health condition or you have significant carer responsibilities, you may require reasonable adjustments to fully access and participate in education. Swinburne's AccessAbility Services can develop an Education Access Plan (EAP) that includes the services and reasonable adjustments that you need.

It is recommended that you register with AccessAbility Services when you first commence your course, but you can contact the service at any time during your studies to find out about reasonable adjustments. Contact [AccessAbility Services](#) to discuss further.

Once you have an approved EAP, you should contact the Unit Convenor as early as possible (e.g. beginning of semester) to discuss your adjustments so that we can better support your learning.

Review of marks

An independent marker reviews all fail grades for major assessment tasks. In addition, a review of assessment is undertaken if your final result is between 45 and 49 or within 2 marks of any grade threshold.

You can ask the Unit Convenor to check the result for an assessment item or your final result. Your request must be made in writing within 10 working days of receiving the result. The Unit Convenor can discuss the marking criteria with you and check the aggregate marks of assessment components to identify if an error has been made. This is known as local resolution. If you are dissatisfied with the outcome of the local resolution, you can lodge a formal complaint.

Feedback, complaints and suggestions

In the first instance, discuss any issues with your Unit Convenor. If your concerns are not resolved or you would prefer not to deal with your Unit Convenor, then you can complete a feedback form. See <https://www.swinburne.edu.au/corporate/feedback/>

Advocacy

Should you require assistance with any academic issues, University statutes, regulations, policies and procedures, you are advised to seek advice from an Academic Student Support Officer at Swinburne Student Life.

For an appointment, please call 19006412 or lodge a query via Student Portal at <https://portal.swin.edu.vn>